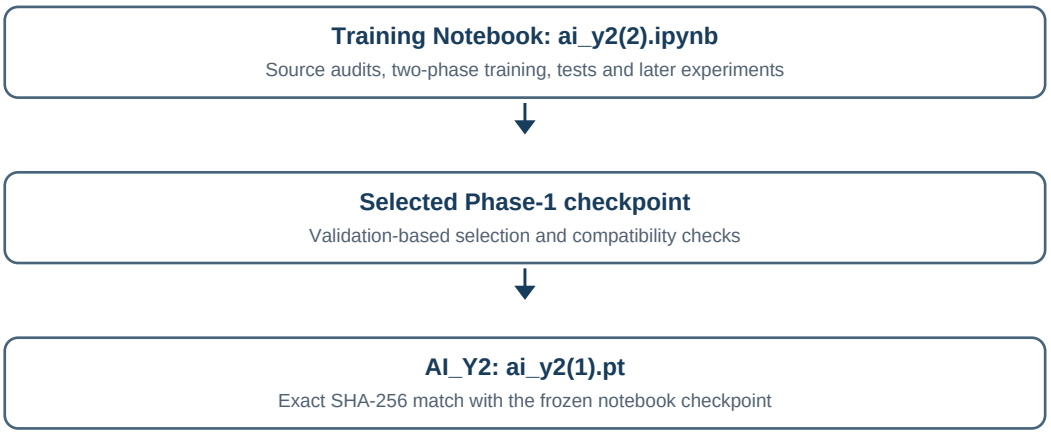
ORALSCAN AI_Y2

YOLO26-M Instance Segmentation
Complete Reverse Engineering, Study & Technical Documentation

TWO SOURCE ARTIFACTS -> ONE VERIFIED MODEL

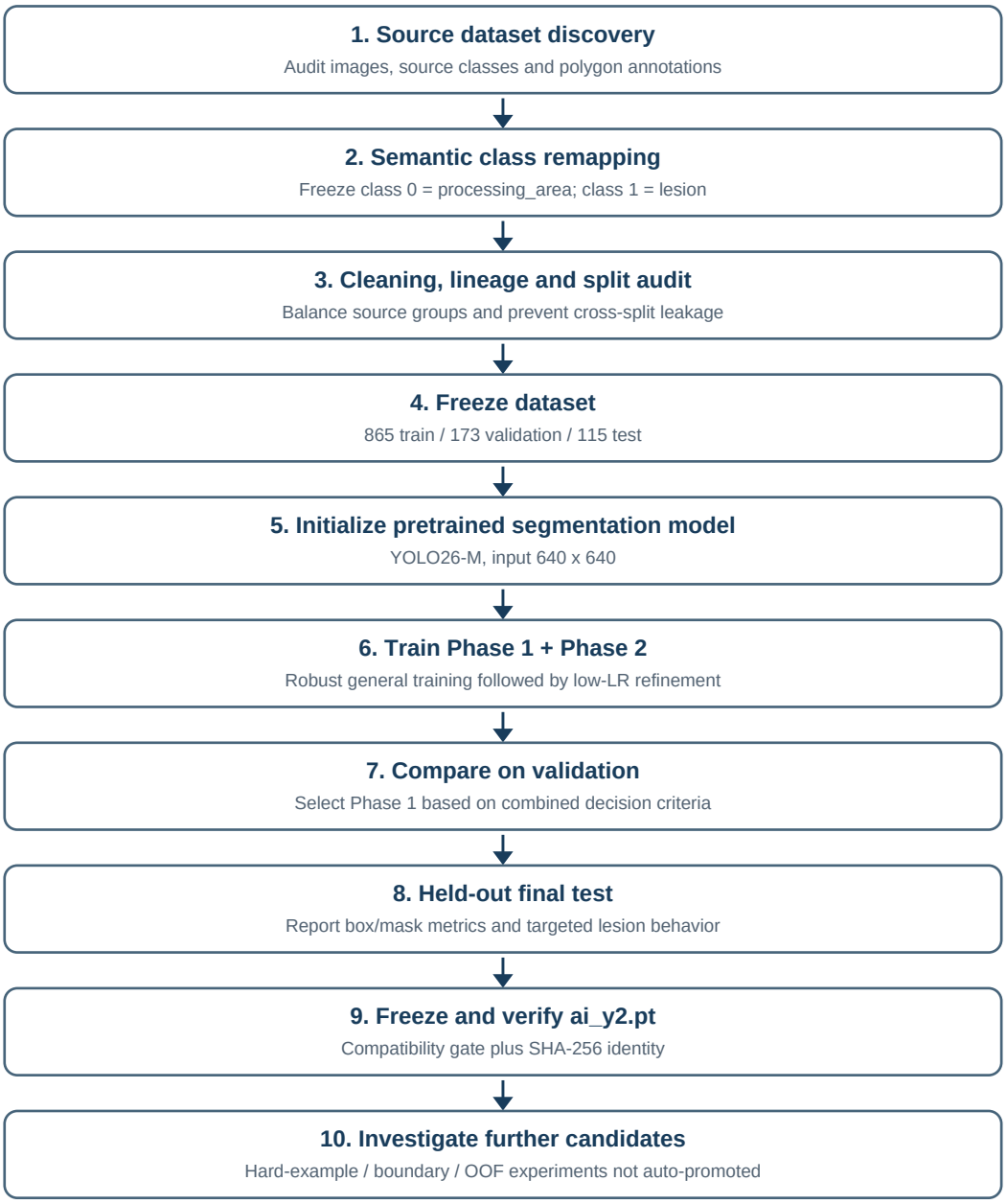


Purpose: long-term study, reproducibility, architecture understanding, deployment review, and integration with OralScan classifier models.

Evidence: uploaded notebook, recorded execution outputs, model archive and AI_Y1 reference document.

AI_Y2 - Complete Training Workflow

DEVELOPMENT AND MODEL SELECTION



Study Note: The OOF-weighted candidate was a separate experimental artifact. The uploaded checkpoint is the frozen Phase-1 selection, not a later automatically promoted candidate.

1. One-Page Quick Reference

Item	Confirmed AI_Y2 configuration
Task	Instance segmentation of oral images
Architecture	YOLO26-M segmentation; yolo26m-seg.yaml; Segment26 head
Uploaded checkpoint	ai_y2(1).pt (same SHA-256 as /content/ai_y2.pt in recorded freeze)
Source notebook	ai_y2(2).ipynb, 23 code cells
Classes	0 = processing_area; 1 = lesion
Image size	640 x 640
Final dataset	1,153 images (865 / 173 / 115)
Phase 1	350 epoch cap, patience 55, LR 0.0004
Phase 2	120 epoch cap, patience 30, LR 0.00010
Training optimizer	AdamW
Mask-target downsampling	mask_ratio=2
Selected checkpoint	Phase-1 best.pt after validation candidate comparison
Fused model summary	146 layers; 23,509,010 parameters; 121.4 GFLOPs
Test mask mAP50 / mAP50-95	0.4835 / 0.2178
Inference	conf=0.25; IoU=0.70; imgsz=640
Primary use	Return boxes, confidences, segmentation masks and polygons for ROI preparation

Study Note: AI_Y2 locates and segments regions. It does not assign the five diagnostic classes of AI_C2 or the four diagnostic classes of AI_C1.

2. Objective and Role in OralScan

AI_Y2 is the spatial front-end of the OralScan vision pipeline. It finds the anatomical processing area and lesion instances in a photograph, then exposes geometry from which the downstream software may create processing-area and lesion-region crops.

• Maintain interchangeable detector class IDs and semantics across Y1 and Y2.

• Keep a processing-area pathway even when a lesion is not predicted.

• Allow the classifier adapter to handle missing detections and a full-image fallback.

3. Dataset Provenance and Frozen Class Contract

The notebook builds a cleaned, source-aware dataset from five source strata. It records group/lineage checks before generating a controlled training/validation/test split. The final dataset directory is /content/AI_Y2_FINAL.

Partition	Images	Processing polygons	Lesion polygons
Train	865	900	787
Validation	173	179	157
Test	115	120	105
Total	1,153	1,199	1,049

3.1 Fixed two-class output mapping

0 = processing_area - annotated usable oral region. **1 = lesion** - annotated lesion region. Cancer, leukoplakia, leukoplakia-like and other disease-specific source labels are not separate YOLO output categories in AI_Y2.

3.2 Quality and leakage controls

• Semantically remap source labels into the two-class vocabulary.

• Validate segmentation polygons and image/label pairing.

• Use source-stratified, group-aware rebalancing for the final split.

• Compare file hashes and lineage groups to detect cross-split contamination.

• Freeze and verify the constructed dataset before training.

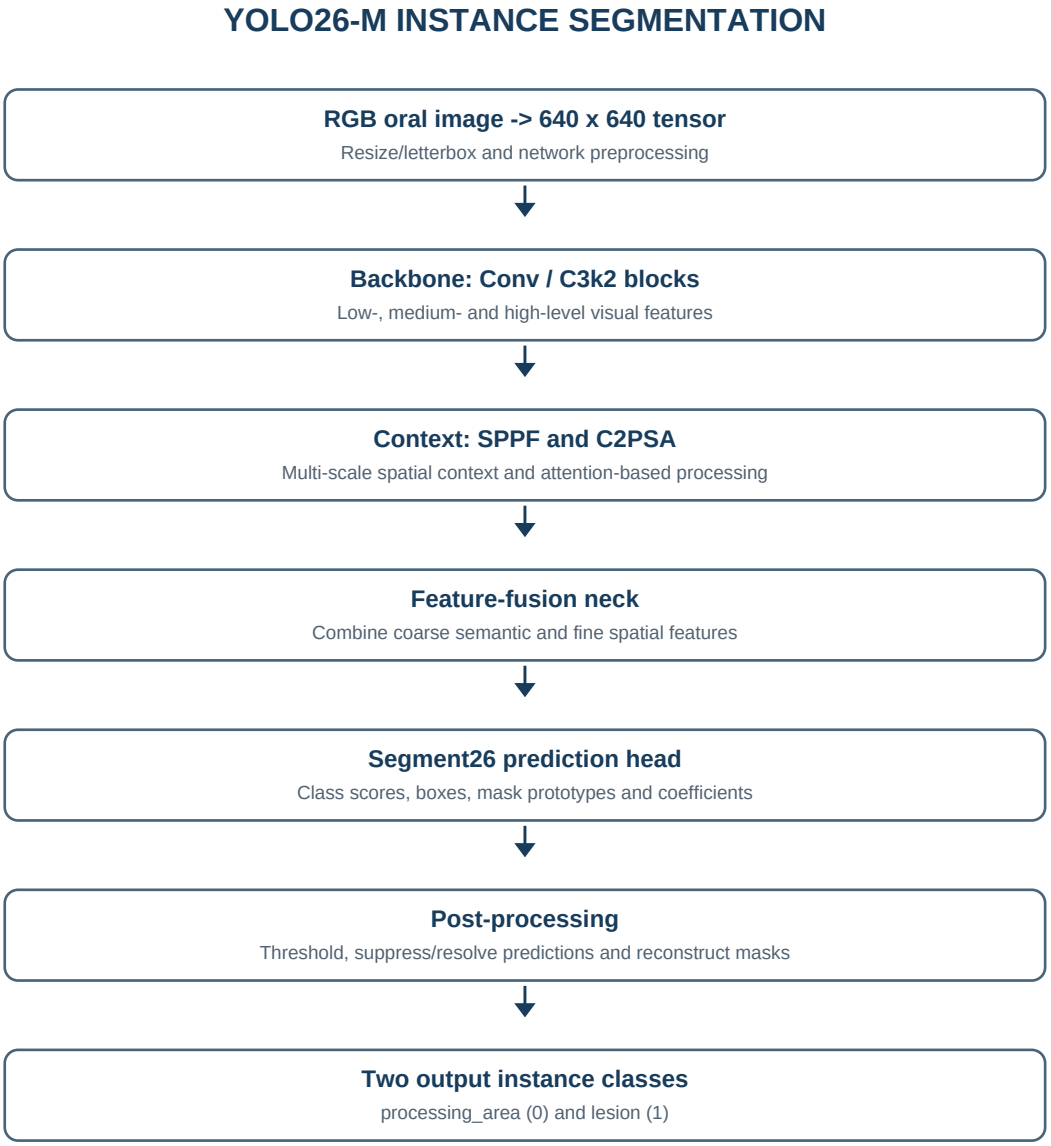
The notebook reports **zero detected lineage cross-split leakage** and **zero exact MD5 cross-split overlap** in its recorded dataset-generation checks. These audits reduce, but cannot eliminate, all near-duplicate or domain-shift concerns.

3.3 Dataset freeze fingerprint

Dataset fingerprint (recorded): `fcf15becb5390bc23d79c81d47e0cf45fe386a74217872c3788ffcf6e3544e6e`

Study Note: Dataset counts are from the notebook outputs; the original raw image archive was not re-evaluated in this report.

4. Reconstructed Neural Network Architecture



This is a functional architecture reconstruction. The model archive names Conv, C3k2, SPPF, C2PSA, Concat and Segment26; it does not by itself establish a complete per-layer graph with every tensor dimension.

5. Network Components Explained

5.1 Convolution and learned features

Convolutions apply trainable kernels over neighboring pixels/features. Earlier representations capture edges, gradients and texture; deeper representations aggregate patterns and contextual evidence needed to localize oral regions.

5.2 C3k2 representation blocks

C3k2-style blocks mix and reuse feature channels efficiently. In a segmentation detector they help learn intermediate visual representations without making the backbone prohibitively expensive.

5.3 SPPF context aggregation

Spatial Pyramid Pooling - Fast (SPPF) aggregates receptive fields so both broad mucosal context and localized features can inform predictions.

5.4 C2PSA and attention

C2PSA introduces attention-based interaction between spatial features. This provides a mechanism for contextual feature refinement; it does not by itself prove medically interpretable attention.

5.5 Multi-scale neck

The neck combines feature maps at different scales. Higher-resolution features help recover local boundaries, while lower-resolution features represent large anatomical structures and semantic context.

5.6 Segment26 head

A segmentation head combines localization/classification predictions with instance-mask representations. In prototype-based formulations, an instance mask is generated from a weighted combination of shared prototype maps followed by post-processing.

Conceptual mask equation: **M_i(x,y) = sigmoid(sum_k [c_(i,k) * P_k(x,y)])**

5.7 Model complexity

Property	Recorded value
Fused layers	146
Fused parameters	23,509,010
Fused compute estimate	121.4 GFLOPs
Task head	Segment26 (segmentation)

6. Training Configuration and Two-Phase Strategy

The notebook uses a pretrained YOLO26-M segmentation checkpoint and trains two controlled candidates. Phase 2 begins from Phase 1 best weights but is not automatically accepted.

Parameter	Phase 1	Phase 2
Goal	Robust general segmentation	Gentler mask/lesion refinement
Epoch limit	350	120
Early stop patience	55	30
Initial LR	0.0004	0.00010
Warmup epochs	4	2
Rotation	8 degrees	3 degrees
Translation	0.04	0.02
Scale	0.12	0.05
Horizontal flip	0.50	0.50
Mosaic	0.10	0
Copy-paste	0	0
Optimizer	AdamW	AdamW
Mask ratio	2	2

6.1 Why two phases?

Phase 1 optimizes a broadly useful feature representation with moderate but clinically controlled augmentation. Phase 2 reduces the learning rate and augmentation strength in an attempt to refine mask boundaries without destroying previously learned localization.

6.2 Why preserve the best checkpoint?

The notebook compares Phase 1 and Phase 2 on held-out validation data rather than assuming that a later training run is superior. The exported model is selected using the validation performance and targeted lesion-detection trade-offs.

7. Validation-Based Candidate Selection

Both candidates were evaluated on the same 173-image validation split. The study measured not only mask mAP but lesion-positive image detection and false lesion predictions in processing-only images.

Validation metric	Phase 1	Phase 2
Mask precision	0.7337	0.7400
Mask recall	0.6393	0.6388
Mask mAP@50	0.5661	0.5586
Mask mAP@50-95	0.2629	0.2649
Lesion mask mAP@50-95	0.2821	0.2577
Processing mask mAP@50-95	0.2436	0.2720
Lesion image recall at conf .25	0.9398	0.8421
False lesion image rate at conf .25	0.1750	0.0750
Notebook selection score	0.575477	0.554232

7.1 Interpreting the choice

Phase 2 achieves marginally higher overall strict mask mAP and fewer false lesion images, but Phase 1 has higher lesion-specific strict mask mAP and substantially stronger lesion-positive image recall at the same threshold. The notebook therefore recommends **Phase 1**.

Study Note: The validation comparison is part of model selection. Final test images must remain outside this decision process to preserve an honest held-out evaluation.

8. Final Held-Out Test Evaluation

The frozen Phase-1 candidate was tested on 115 independent held-out images with 225 annotated instances in total.

Overall test metric	Value
Mask precision	0.6550
Mask recall	0.5940
Mask mAP@50	0.4835
Mask mAP@50-95	0.2178
Box precision	0.8192
Box recall	0.7270
Box mAP@50	0.6960
Box mAP@50-95	0.4247

8.1 Per-class mask results

Class	Images / instances	Mask P	Mask R	mAP50	mAP50-95
processing_area	115 / 120	0.688	0.683	0.550	0.254
lesion	89 / 105	0.622	0.505	0.417	0.182

8.2 Lesion image presence behavior (conf = 0.25)

Measure	Recorded test finding
Lesion-positive images	89
Correctly flagged lesion-positive images	82 (92.13% image-level recall)
Processing-only images	26
Processing-only images falsely flagged as lesions	2 (7.69% image-level false lesion rate)
Processing-area presence rate	0.9826

Image-level lesion presence recall is not equivalent to box recall or mask recall. A weakly overlapping instance may still count as a detected lesion-present image.

9. Evaluation Mathematics and Failure Modes

9.1 Precision and recall

Precision = TP / (TP + FP) - proportion of positive predictions that are correct. **Recall = TP / (TP + FN)** - proportion of ground-truth positives recovered.

9.2 Intersection over Union

IoU = area(predicted region intersection ground truth) / area(predicted region union ground truth). Higher overlap means closer agreement with the annotated location or contour.

9.3 mAP thresholds

mAP@50 summarizes average precision with overlap threshold 0.50. mAP@50-95 averages AP over increasingly strict overlap thresholds, making it more sensitive to poor box placement and mask boundaries.

9.4 Why lesion mask quality is lower

The test report shows lesion mask mAP@50-95 of 0.1817 compared with 0.2539 for the processing area. Plausible technical causes include small lesion size, inconsistent contours, visually ambiguous boundaries and source variation, but the notebook metrics alone do not isolate the cause.

9.5 Common failure modes to audit

• Missed lesion despite successful processing-area detection.

• False lesion predicted on an image with no lesion annotation.

• Bounding box overlaps target but mask contour is inaccurate.

• Multiple competing lesion instances and inaccurate confidence ranking.

• Mismatch between training annotations and real-world image conditions.

Study Note: These quantitative results are research/test-set performance figures, not clinical sensitivity/specificity or evidence of generalization across all devices and populations.

10. Inference Workflow and Classifier Handoff

The notebook verifies the presence of boxes.cls, boxes.conf, boxes.xyxy, masks.data and masks.xy on a detection-bearing example. In general, masks can be absent for images with zero predictions and must be checked before access.

11. Checkpoint Identity and Additional Research

11.1 Exact checkpoint identity

SHA-256 for uploaded ai_y2(1).pt and the frozen model recorded in the notebook:

```
6c80c9a9449dd99362aba69c1a2e18e71619b38ffad53478b6211b0f20cf4b93
```

This exact match ties the submitted binary checkpoint to the final Phase-1 model selected and evaluated in the notebook.

11.2 Post-freeze improvement work

• Validation failure analysis and annotation geometry review.

• Training-only hard-example discovery and weighting.

• Boundary-focused refinement candidate training.

• True five-fold out-of-fold (OOF) training and hard-case mining.

• 1,210-entry OOF-weighted training candidate with separate validation.

The notebook explicitly reports **"Current ai_y2.pt overwritten: NO"** for the OOF-weighted candidate. Its validation study is an experiment rather than the deployment model used for the test results above.

11.3 Runtime safety and verification

A .pt archive may include serialized Python objects. Load it only from trusted sources using a compatible PyTorch/Ultralytics environment. For deployment, verify successful deserialization, inference shapes, class names, confidence semantics, mask presence checks, and actual runtime latency.

12. Reproducibility Checklist

• Acquire the original labeled source datasets and their YAML files.

• Run source audits, class remapping, polygon verification and lineage grouping.

• Freeze dataset with the same 865 / 173 / 115 split and fingerprint.

• Maintain {0: processing_area, 1: lesion} exactly.

• Use the YOLO26-M segmentation backbone with 640-pixel input and mask_ratio=2.

• Repeat both phase configurations and preserve their best.pt checkpoints.

• Choose the candidate using validation, not test metrics.

• Evaluate once on the held-out test with documented thresholds.

• Confirm the final model SHA-256 and expected Ultralytics prediction schema.

• Validate results on additional unseen clinical sources before real-world use.

13. Study Summary

Question	Answer
What is AI_Y2?	One YOLO26-M oral instance segmentation detector.
How many classes?	Two: processing_area and lesion.
How much final data?	1,153 images; 865 train, 173 validation, 115 test.
Which checkpoint won?	Phase 1 selected on validation.
What does it output?	Class IDs, confidences, boxes, masks and polygons.
Does it diagnose disease?	No. That is performed by a downstream classifier.
Were later OOF candidates promoted?	No; kept separate in the supplied notebook.

14. Source Provenance and Scope

Primary sources: uploaded ai_y2(2).ipynb with recorded executions, uploaded ai_y2(1).pt verified by SHA-256, and the uploaded AI_Y1 Doc.docx as the layout/organization reference. Numeric values reflect notebook records; no fresh training or test-set evaluation was performed. Network block descriptions are explanatory interpretations, distinct from explicitly extracted checkpoint metadata.

Document prepared for engineering study and OralScan research documentation.